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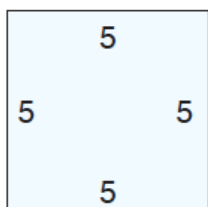
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Using a Given Formula

Formulas are useful tools for solving certain kinds of problems. A **formula** is an equation that shows a constant relationship between variables.

For example, you have already learned that **perimeter** is the total distance around a figure. You can find the perimeter of any figure by adding its sides. The perimeter of the square in the drawing is 20 because $5 + 5 + 5 + 5 = 20$.



From the example, you can write directions stating how to find the perimeter of a square: *Take 1 side and multiply it by 4.*

You can write those directions as a formula. Let P represent perimeter and s represent a side. Write $P = 4s$.

On the GED® Math Test, you are expected to know many basic formulas. Good test takers memorize these basic formulas.

When you take the GED Math Test, you will be able to access a formula sheet by clicking on a link during the test. If you need any other formula to answer a question during the test, you will be given it as part of the question.

To use a formula, substitute the given values and use the order of operations to find the value that you don't know.

Example

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Barbara borrowed \$3,000. She financed the loan using simple interest. She agreed to pay a rate of 4% interest for 5 years. How much interest will she pay on the loan?

simple interest: $I = prt$

The variable p represents principal (the amount borrowed), r represents rate, and t represents time in years.

Step 1

Substitute the given values in the formula.

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Using a Given Formula

Solve.
 $I = \$3,000 \times 0.04 \times 5 = \600
The interest (I) is \$600.
Barbara will pay **\$600** in interest on the loan.

📌 Formulas

Hide

Perimeter (P)	Formula	Definition
Rectangle	$P = 2l + 2w$	l = length, w = width
Polygon with n sides	$P = s_1 + s_2 + \dots + s_n$	s_1 , etc. = each side
Circle	$C = \pi d$ or $C = 2\pi r$	C = Circumference, d = diameter, r = radius

Area (A)	Formula	Definition
Rectangle	$A = lw$ or $A = bh$	l = length, w = width, b = base, h = height
Square	$A = s^2$	s = side
Parallelogram	$A = bh$	b = base, h = height
Triangle	$A = \frac{1}{2}bh$	b = base, h = height
Trapezoid	$A = \frac{1}{2}h(b_1 + b_2)$	b_1 and b_2 = bases, h = height
Circle	$A = \pi r^2$	r = radius

Volume (V)	Formula	Definition
Rectangular/right prism (includes cube)	$V = Bh$	B = area of base, h = height
Cylinder	$V = \pi r^2 h$	r = radius, h = height
Pyramid	$V = \frac{1}{3}Bh$	B = area of base, h = height

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Using a Given Formula



	$\pi r^2 h$	height
Sphere	$V = \frac{4}{3} \pi r^3$	$r = \text{radius}$

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Key Term
 perimeter

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